

SportRise

Discover. Train. Rise.

FUNCTIONAL DESIGN DOCUMENT

FDD — Based on Standard FDD Template

Architecture | ERD | UI Flows | APIs | Business Rules | Test Scenarios
sportrise.in | Phase 1 — Football, Hyderabad | Version 1.0 | June 2026

Document Type	Derived From	Prepared By	Date
FDD v1.0	BRD v1.0 + FRD v1.0	Business Analyst + Tech Lead	June 2026

Digital Edify BA & AI Course — SportRise Project | sportrise.in

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1. INTRODUCTION

1.1 Purpose and Scope

This Functional Design Document (FDD) provides the complete technical and functional specification for SportRise Phase 1. It translates the business requirements defined in the BRD v1.0 and the functional requirements in the FRD v1.0 into a detailed design specification that developers, testers, and architects can use to build, test, and validate the system.

SportRise Phase 1 covers the Android mobile application for football athletes in Hyderabad, including student registration, coach marketplace, tournament ecosystem, XP engine, National Prospect badge system, and scout dashboard.

1.2 Related Documents

Document	Version	Description
Business Requirements Document (BRD)	v1.0	Defines what the business needs from SportRise Phase 1
Functional Requirements Document (FRD)	v1.0	Defines how the system must behave at functional level
UI Design Document	v1.0	Wireframes and screen specifications for all 36 screens
UAT Document	v1.0	UAT scripts, defect log, regression suite, and sign-off
SQL Reference Document	v1.0	Database queries for BA validation and reporting
Competitive Analysis Document	v1.0	Market positioning and competitor strategy

1.3 Assumptions and Constraints

Assumptions

- All coaches have valid certifications from recognized bodies available for upload as PDF documents
- Students in Hyderabad have Android 8.0+ smartphones with 4G internet connectivity
- Tournament organizers will adopt digital processes when registration is significantly easier than paper
- SMS gateway provider delivers OTPs within 60 seconds on standard Airtel, Jio, and BSNL SIMs
- Google Play Store will approve the app within 3-7 working days of submission

Constraints

Constraint	Value	Impact on Design
Platform	Android only — minimum Android 8.0 (API Level 26)	iOS features deferred to Phase 2
Budget	Phase 1 maximum Rs 15-20 lakhs	Technology stack must use cost-effective or free tier services
Timeline	MVP by September 2026	Feature scope is tightly managed — must avoid scope creep
Geography	Hyderabad only	Location-based features limited to one city in Phase 1

Sport	Football only	Sport-specific features (e.g. positions) designed for football first
Data Residency	India servers only — DPDP Act 2023	AWS Mumbai region or equivalent India-based hosting

1.4 Stakeholders and Roles

Role	Stakeholder	FDD Responsibility
Business Analyst	BA Team	Owns FDD. Reviews for completeness and traceability to BRD and FRD
Technical Lead	Development Team	Reviews architecture sections. Signs off on technical feasibility
Backend Developer	Development Team	Implements API specifications, business logic, and data model
Frontend Developer	Development Team	Implements UI specifications, navigation flows, and screen layouts
QA Lead	QA Team	Derives test cases from Section 10. Validates acceptance criteria
Product Owner	Business Team	Reviews functional requirements and use cases. Approves scope
Admin Team	Operations	Reviews admin module specifications and workflow rules

2. SOLUTION OVERVIEW

2.1 High-Level Solution Architecture

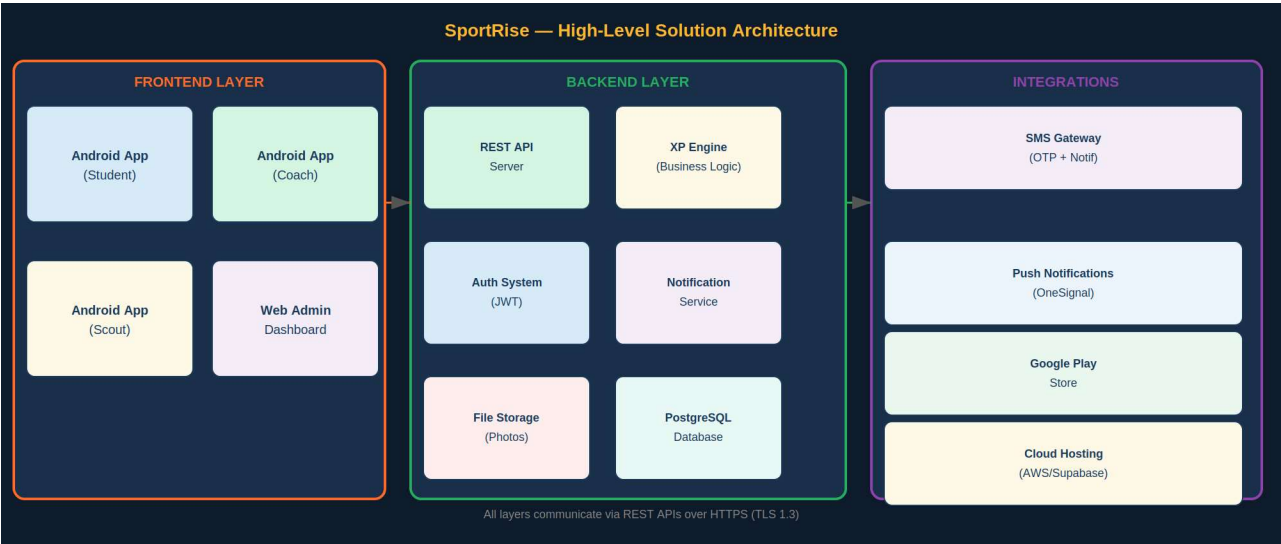


Figure 2.1 — SportRise High-Level Solution Architecture (3-Layer: Frontend, Backend, Integrations)

2.2 System Components

Component	Type	Description
Android App (Student)	Mobile Frontend	Student-facing app: registration, coach discovery, tournament, XP dashboard
Android App (Coach)	Mobile Frontend	Coach-facing app: profile management, session rating, student tracking
Android App (Scout)	Mobile Frontend	Scout-facing app: athlete search, profile view, shortlist management
Web Admin Dashboard	Web Frontend	Admin panel: coach verification queue, tournament approval, platform analytics
REST API Server	Backend	Node.js/FastAPI server handling all business logic and data operations
XP Engine	Backend Service	Dedicated business logic service for XP calculation, level management, NP badge award
Auth System	Backend Service	JWT-based authentication with OTP verification and parental consent flow
Notification Service	Backend Service	Coordinates push notifications (OneSignal) and SMS (MSG91) delivery
PostgreSQL Database	Data Layer	Primary relational database storing all platform data
File Storage	Data Layer	AWS S3 or Cloudinary for profile photo and certification document storage

2.3 Technology Stack

Layer	Technology	Justification
Mobile Frontend	React Native (or FlutterFlow for no-code MVP)	Single codebase for Android. FlutterFlow for rapid Phase 1 MVP.
Backend API	Node.js with Express or Python with FastAPI	Covered in Digital Edify BA & AI course curriculum
Database	PostgreSQL	Covered in course. ACID compliance for XP transaction integrity.
Authentication	JWT (JSON Web Tokens)	Industry standard. Stateless. Scalable.
File Storage	AWS S3 or Cloudinary	Free tier adequate for Phase 1 photo storage volumes
Push Notifications	OneSignal (free tier)	Supports up to 10,000 devices free. Easy Android integration.
SMS Gateway	MSG91 or Twilio	OTP delivery across all major Indian operators
Cloud Hosting	AWS Mumbai Region or Supabase	India data residency compliance. Supabase free tier for MVP.
Admin Dashboard	React.js web app	Simple web interface for admin verification workflows
Analytics	Firebase Analytics (free)	Crash reporting, DAU tracking, funnel analysis

2.4 Integration Landscape



Figure 2.2 — SportRise API Architecture and Integration Landscape

3. FUNCTIONAL REQUIREMENTS DETAIL

3.1 Functional Requirements Matrix

FR ID	Module	Requirement	Priority	BRD Ref
FR-REG-001	Registration	Multi-step registration form with progress bar (Steps 1-3)	Must Have	BR-001
FR-REG-002	Registration	OTP verification to mobile number within 60 seconds	Must Have	BR-001
FR-REG-003	Registration	Parental consent flow for users aged 5-17	Must Have	BR-012
FR-REG-004	Registration	Profile photo upload (JPG/PNG, max 5MB, cropped 1:1)	Should Have	BR-001
FR-COA-001	Coach	Coach discovery with sport and city filters	Must Have	BR-002
FR-COA-002	Coach	Full coach profile with certifications, ratings, fee, availability	Must Have	BR-002
FR-COA-003	Coach	Session booking with calendar date and time slot selection	Must Have	BR-002
FR-COA-004	Coach	Coach session rating (1-5) with exceptional flag	Must Have	BR-004
FR-TRN-001	Tournament	Tournament discovery with sport filter and registration status	Must Have	BR-003, BR-010
FR-TRN-002	Tournament	One-tap tournament registration with auto XP award (20 XP)	Must Have	BR-003
FR-TRN-003	Tournament	Organizer result entry triggering XP calculation	Must Have	BR-004
FR-XPE-001	XP Engine	Automatic XP calculation per XP Rules Table	Must Have	BR-004
FR-XPE-002	XP Engine	Level-up detection and celebration notification	Must Have	BR-005
FR-NPS-001	NP System	NP badge award when any trigger condition is met	Must Have	BR-006
FR-NPS-002	NP System	Scout dashboard National Prospects section update within 60 min	Must Have	BR-006, BR-008
FR-SCT-001	Scout	Athlete search with sport, city, age, level filters	Must Have	BR-008
FR-SCT-002	Scout	Full athlete profile view with complete performance history	Must Have	BR-008
FR-ADM-001	Admin	Coach verification queue with approve/reject workflow	Must Have	BR-011

FR-ADM-002	Admin	Tournament verification queue with approve/reject workflow	Must Have	BR-011
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3.2 Use Cases

UC ID	Use Case Name	Primary Actor	Trigger	Outcome
UC-01	Register as Student Athlete	Student	Taps GET STARTED on Welcome Screen	Student profile created, OTP verified, dashboard accessible
UC-02	Find and Book a Coach	Student	Taps Find Coach on dashboard	Coaching session booked, confirmation sent, appears in My Bookings
UC-03	Register for Tournament	Student	Taps REGISTER on tournament card	Registration confirmed, 20 XP awarded, appears in My Registrations
UC-04	Rate Student After Session	Coach	Taps Rate for completed session	XP awarded to student, level-up checked, NP checked
UC-05	Submit Tournament Results	Organizer	Enters result after tournament event	XP awarded based on result, NP conditions re-evaluated
UC-06	Search for Top Athletes	Scout	Logs in and uses search filters	Ranked list of verified athletes matching filters returned
UC-07	View Athlete Profile	Scout	Taps on athlete card in search results	Full performance history shown, student notified within 15 min
UC-08	Verify Coach Profile	Admin	Receives verification request in queue	Coach approved or rejected with reason, coach notified
UC-09	Earn National Prospect Badge	System + Student	NP trigger condition met after any XP event	NP badge awarded, scout dashboard updated, student congratulated
UC-10	Send Recruitment Interest	Scout	Taps SEND INTEREST on athlete profile	Interest recorded, student receives notification

3.3 User Workflows

Student Coaching Workflow — TO-BE Process

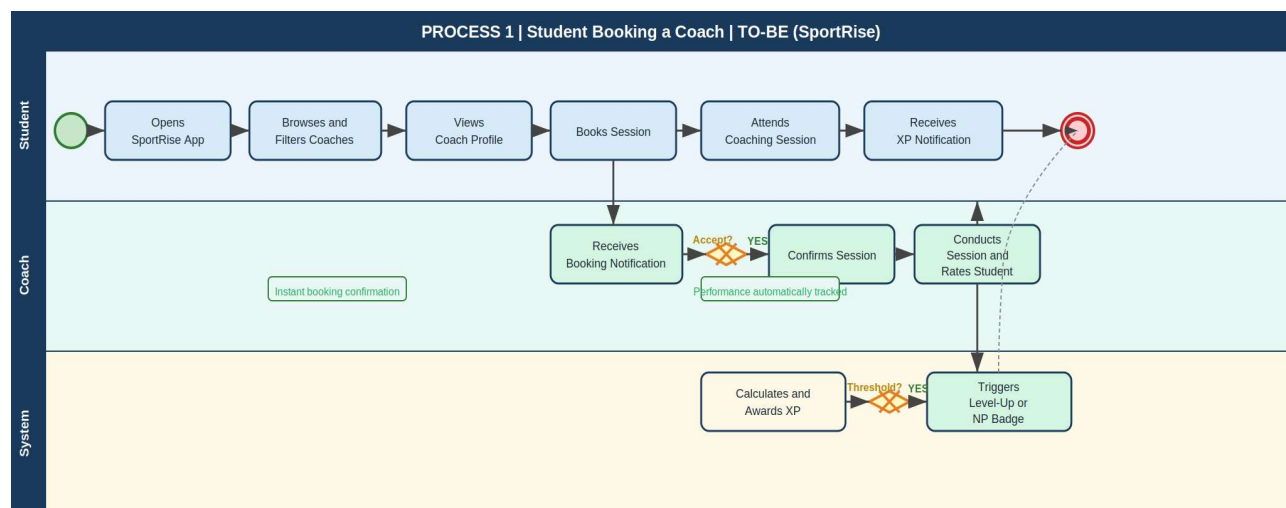


Figure 3.1 — Student books coach and earns XP (TO-BE BPMN Process Flow)

National Prospect Identification Workflow

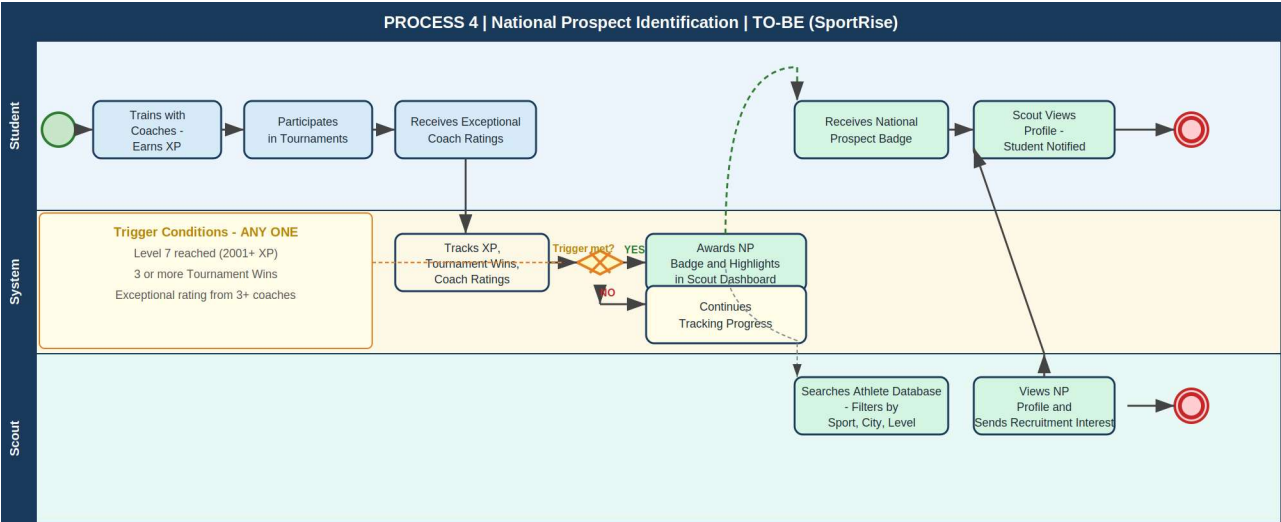


Figure 3.2 — National Prospect automatic identification (TO-BE BPMN Process Flow)

3.4 Business Rules Engine

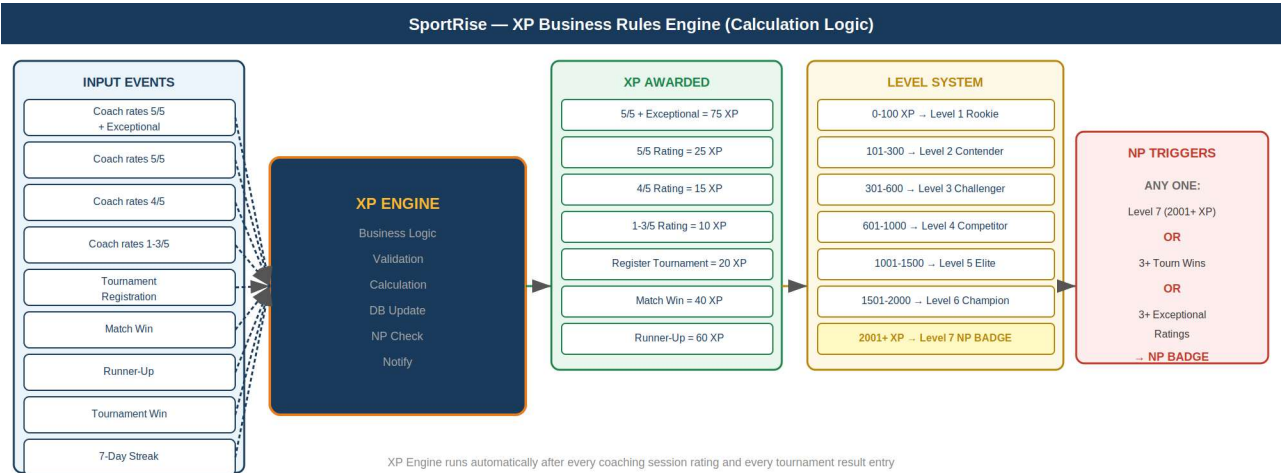


Figure 3.3 — XP Business Rules Engine: Inputs, Calculation Logic, Level System, and NP Triggers

4. USER INTERFACE SPECIFICATIONS

4.1 Screen Layouts and Mockups

SportRise Phase 1 requires 36 screens across 5 user roles. Priority 1 screens (10 screens) are the core flows that must be designed and built first. Full wireframe specifications are in the UI Design Document v1.0. Key screen specifications are summarized below.

ID	Screen Name	User Role	Priority	Key Elements
S-01	Welcome Screen	All Users	Must Have	Logo, tagline, GET STARTED button, login link
S-02	Role Selection	All Users	Must Have	4 role cards: Student, Coach, Organizer, Scout
S-03	Student Registration Step 1	Student	Must Have	Name, age, city dropdown, mobile with +91
S-04	Student Registration Step 2	Student	Must Have	Sport selection grid (2-col, icons)
S-05	OTP Verification	All Users	Must Have	6-box OTP input, 60-sec resend timer
S-06	Parental Consent	Under-18	Must Have	Explanation, parent mobile field, consent OTP
S-07	Student Dashboard	Student	Must Have	Level banner, XP bar, quick actions, upcoming sessions
S-08	Coach Discovery	Student	Must Have	Search bar, filter chips, verified coach cards
S-09	Coach Profile	Student	Must Have	Photo, name, sport, rating, fees, BOOK button
S-10	XP and Level Screen	Student	Must Have	Level journey, XP bar, earn more section
S-11	Scout Dashboard	Scout	Must Have	NP section, search bar, filter controls, results
S-12	Athlete Profile (Scout view)	Scout	Must Have	Full performance history, coaching ratings, tournament wins

4.2 Navigation Flow

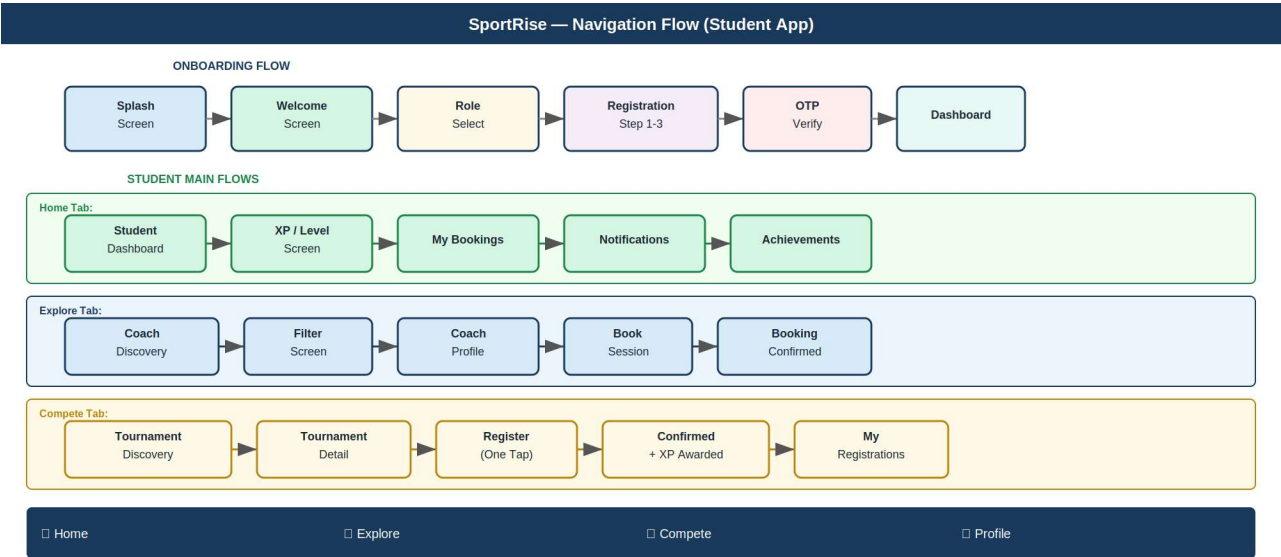


Figure 4.1 — SportRise Student App Navigation Flow (Onboarding + 3 Main Tabs)

4.3 Field Definitions

Student Registration Form — Step 1

Field	Type	Required	Max Length	Validation Rule
Full Name	Text input	Yes	60 chars	Min 2 chars, letters and spaces only, no special characters
Age	Numeric input	Yes	2 digits	Min 5, Max 30, integers only
City	Dropdown select	Yes	N/A	Must select from pre-populated list — free text not allowed
Mobile Number	Numeric input	Yes	10 digits	Exactly 10 digits after +91 prefix, numeric only
Profile Photo	Image upload	No	5 MB	JPG or PNG only, minimum 200x200px, cropped to 1:1

Coach Search Filters

Filter	Type	Default	Options / Range
Sport	Dropdown	Football (pre-filled from student profile)	All sports in database — dropdown list
City	Dropdown	Hyderabad (pre-filled from student profile)	All cities in database — dropdown list
Price Range	Range slider	Rs 0 to Rs 10000	Rs 100 to Rs 10,000 in Rs 100 increments
Minimum Rating	Star selector	Any	1 star to 5 stars in 0.5 increments

Available Today	Toggle switch	Off	On / Off — filters to coaches available today
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4.4 UI/UX Guidelines

Guideline	Specification
Primary Color	Deep Navy Blue #1A3A5C — used for headers, primary text, pool headers
Accent Color	Electric Orange #FF6B2B — used for all primary CTA buttons, XP bar, active navigation icon
Success Color	Green #27AE60 — verified badges, level-up states, success toasts
Error Color	Red #C0392B — form validation errors, rejection states, failed actions
Primary Font	Poppins Bold for headings (H1: 24sp, H2: 20sp). Inter Regular for body text (16sp)
Touch Targets	All interactive elements minimum 48x48dp — accessibility requirement
Spacing System	All spacing in multiples of 8dp: 8, 16, 24, 32, 48, 64
Loading State	Skeleton screens for all data-loading states — never show blank screen or spinner only
Empty State	Every list screen has a designed empty state with illustration and actionable message
Error State	Every action that can fail shows a human-friendly error with a retry or back action
Bottom Navigation	4 tabs: Home, Explore, Compete, Profile — icons with text labels — 64dp height
One Primary Action	Every screen has exactly ONE primary orange CTA button — no competing calls to action

5. DATA SPECIFICATIONS

5.1 Data Model (ERD)

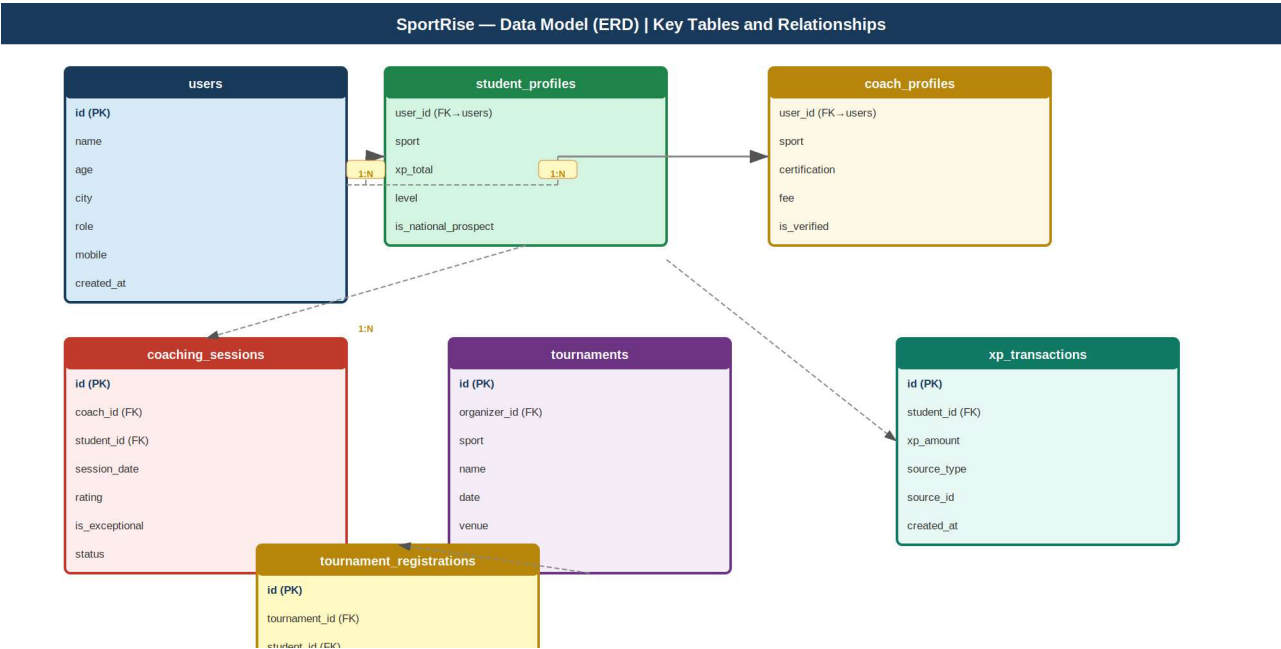


Figure 5.1 — SportRise Entity Relationship Diagram (ERD) — Key Tables and Relationships

5.2 Object Definitions

Table	Primary Key	Description
users	id (UUID)	Master user table for all roles — student, coach, scout, organizer, admin
student_profiles	user_id (FK → users.id)	Extended profile for student athletes — sport, XP, level, NP status
coach_profiles	user_id (FK → users.id)	Extended profile for coaches — sport, certification, fee, verified status
coaching_sessions	id (UUID)	All coaching sessions — coach, student, date, rating, exceptional flag, status
tournaments	id (UUID)	Tournament listings — organizer, sport, name, date, venue, verified status
tournament_registrations	id (UUID)	Student tournament registrations and results
xp_transactions	id (UUID)	Immutable audit trail of every XP point earned by every student
notifications	id (UUID)	All in-app and push notifications sent to users
scout_shortlists	id (UUID)	Scout saved athlete shortlists
parental_consents	id (UUID)	Parental consent records for users under 18

5.3 Field Mappings — Key Tables

student_profiles table

Column	Data Type	Nullable	Description
user_id	UUID	NOT NULL	Foreign key to users.id — one-to-one relationship
sport	VARCHAR(50)	NOT NULL	Primary sport of the student — e.g. Football, Cricket
sport_ids	INTEGER[]	NOT NULL	Array of all sport IDs selected during registration
xp_total	INTEGER	NOT NULL DEFAULT 0	Running total of all XP earned — updated by XP Engine
level	INTEGER	NOT NULL DEFAULT 1	Current level (1-7) — derived from xp_total
is_national_prospect	BOOLEAN	NOT NULL DEFAULT false	True when any NP trigger condition is met
np_trigger	VARCHAR(20)	NULLABLE	Records which trigger: XP_THRESHOLD, WINS, EXCEPTIONAL
np_awarded_at	TIMESTAMP	NULLABLE	Timestamp when NP badge was first awarded
profile_complete	BOOLEAN	NOT NULL DEFAULT false	True when all 3 registration steps are complete
city	VARCHAR(100)	NOT NULL	Student's city — used for coach and scout filtering

5.4 Data Validation Rules

Rule ID	Field / Table	Validation Rule
DVR-01	users.age	CHECK (age >= 5 AND age <= 30) — enforced at DB constraint level
DVR-02	users.mobile	UNIQUE constraint — no two accounts with same mobile number
DVR-03	coach_profiles.fee	CHECK (fee >= 100 AND fee <= 10000) — fee in valid range
DVR-04	coaching_sessions.rating	CHECK (rating >= 1 AND rating <= 5) — valid star rating
DVR-05	xp_transactions.xp_amount	CHECK (xp_amount > 0) — XP must always be positive
DVR-06	student_profiles.level	CHECK (level >= 1 AND level <= 7) — valid level range
DVR-07	student_profiles.xp_total	CHECK (xp_total >= 0) — XP cannot be negative
DVR-08	tournament_registrations	UNIQUE (tournament_id, student_id) — no duplicate registrations
DVR-09	coaching_sessions	coach_id != student_id — coach cannot rate themselves
DVR-10	parental_consents.parent_mobile	parent_mobile != student_mobile — must be different numbers

6. INTEGRATION SPECIFICATIONS

6.1 Integration Points

Integration	Direction	Protocol	Purpose
SMS Gateway (MSG91)	Outbound from SportRise	REST API over HTTPS	OTP delivery for registration, login, and parental consent
Push Notifications (OneSignal)	Outbound from SportRise	REST API over HTTPS	XP notifications, level-up, NP badge, scout view alerts
Google Play Store	Outbound from Dev Team	Play Console API	App distribution and update delivery to Android devices
AWS S3 / Cloudinary	Bi-directional	REST API over HTTPS	Upload and retrieve profile photos and coach certification documents
Firebase Analytics	Outbound from App	Firebase SDK	Crash reporting, DAU tracking, funnel analysis, event tracking
Email Service (SendGrid)	Outbound from SportRise	REST API over HTTPS	Coach approval/rejection emails, registration confirmation emails

6.2 API Specifications — Key Endpoints

```
# AUTH MODULE
POST /api/v1/auth/register
Body: { name, age, city, mobile, role, sport_ids[] }
Response: { user_id, status: 'OTP_SENT' }

POST /api/v1/auth/verify-otp
Body: { mobile, otp, user_id }
Response: { jwt_token, user_profile, redirect_to }

POST /api/v1/auth/parental-consent
Body: { student_id, parent_mobile }
Response: { status: 'CONSENT_OTP_SENT' }

# COACH MODULE
GET /api/v1/coaches
Query: ?sport=Football&city=Hyderabad&min_rating=4&max_fee=600
Response: { coaches: [], total: 0, page: 1 }

GET /api/v1/coaches/:id
Response: { coach_profile, sessions, reviews }

POST /api/v1/sessions
Body: { coach_id, date, time_slot }
Response: { session_id, status: 'CONFIRMED', booking_details }

PATCH /api/v1/sessions/:id/rate
Body: { rating, is_exceptional, note }
Response: { xp_awarded, new_xp_total, new_level, np_awarded }

# SCOUT MODULE
```

```
GET      /api/v1/scouts/search
Query:   ?sport=Football&city=Hyderabad&level_min=5&age_max=20
Response: { athletes: [], np_athletes: [], total: 0 }
```

6.3 Data Exchange Formats

All API requests and responses use JSON format with UTF-8 encoding. All timestamps are in ISO 8601 format (YYYY-MM-DDTHH:MM:SSZ). All monetary values are in Indian Rupees (INR) as integers (no decimal).

```
# Sample Athlete Profile API Response
{
  "student_id": "uuid-here",
  "name": "Ravi Teja",
  "age": 17,
  "city": "Hyderabad",
  "sport": "Football",
  "level": 7,
  "level_name": "National Prospect",
  "xp_total": 2340,
  "is_national_prospect": true,
  "np_trigger": "XP_THRESHOLD",
  "coaching_history": [
    { "coach": "Suresh Nair", "date": "2026-05-15", "rating": 5, "xp": 25 }
  ],
  "tournament_history": [
    { "tournament": "Hyderabad Cup", "result": "winner", "xp": 100 }
  ],
  "total_tournament_wins": 3
}
```

6.4 Error Handling

HTTP Code	Error Type	Handling
400	Bad Request	Return field-level validation errors in JSON. Frontend shows inline error messages.
401	Unauthorized	JWT token missing or expired. Frontend redirects to login screen.
403	Forbidden	User role does not have permission. Show: 'You do not have access to this feature.'
404	Not Found	Resource does not exist. Show friendly 'not found' message with back button.
409	Conflict	Duplicate action (e.g. already registered for tournament). Show specific conflict message.
429	Rate Limited	Too many OTP requests. Show: 'Too many attempts. Please wait 30 minutes.'
500	Server Error	Internal error. Show: 'Something went wrong. Please try again.' Log internally.
503	Service Unavailable	System maintenance. Show maintenance screen with estimated return time.

7. BUSINESS RULES

7.1 Validation Rules

ID	Rule Name	Validation Logic
VR-01	Age Range	User age must be ≥ 5 AND ≤ 30 . Checked at form submission AND API level. DB constraint enforced.
VR-02	Mobile Uniqueness	Mobile number must be unique across all user accounts. API returns 409 if duplicate.
VR-03	Parent Mobile Different	Parent consent mobile must $\neq$ student mobile. Frontend and API both validate.
VR-04	OTP Expiry	OTP expires after exactly 10 minutes. System checks OTP creation timestamp on each attempt.
VR-05	OTP Attempt Limit	Maximum 3 failed OTP attempts triggers 30-minute lockout. Counter resets on successful verification.
VR-06	Coach Verification Gate	coach_profiles.is_verified must = true before ANY coach data is returned in student-facing API calls.
VR-07	Tournament Deadline	tournament_registrations can only be created when tournaments.registration_deadline > NOW().
VR-08	Unique Registration	UNIQUE(tournament_id, student_id) DB constraint. API returns 409 on duplicate registration attempt.
VR-09	Immutable XP	xp_transactions records have no UPDATE or DELETE endpoints. Corrections require new compensating transactions.
VR-10	NP Distinct Coaches	NP Condition 3 check counts DISTINCT coach_id values. Same coach exceptional ratings do not count multiple times.

7.2 Calculation Logic

Calc ID	Calculation	Formula / Logic	Trigger
CALC-01	XP from coaching rating	IF rating=5 AND is_exceptional THEN 75 ELIF rating=5 THEN 25 ELIF rating=4 THEN 15 ELSE 10	Coach submits rating
CALC-02	XP from tournament result	IF result=winner THEN 100 ELIF result=runner_up THEN 60 ELIF result=match_win THEN 40	Organizer enters result
CALC-03	XP from registration	20 XP awarded once per tournament per student	Student registers for tournament
CALC-04	XP from streak	30 XP when student has sessions on 7 consecutive calendar days	Session completed — streak counter updated

CALC-05	Level from XP	CASE WHEN xp<100 THEN 1 WHEN xp<300 THEN 2 WHEN xp<600 THEN 3 WHEN xp<1000 THEN 4 WHEN xp<1500 THEN 5 WHEN xp<2000 THEN 6 ELSE 7	After every XP update
CALC-06	NP Condition 1	is_national_prospect = TRUE WHEN xp_total >= 2001	After every XP update
CALC-07	NP Condition 2	is_national_prospect = TRUE WHEN COUNT(results WHERE result=winner) >= 3	After tournament result entry
CALC-08	NP Condition 3	is_national_prospect = TRUE WHEN COUNT(DISTINCT coach_id WHERE is_exceptional=true) >= 3	After coach rating with exceptional

7.3 Workflow Rules

Coach Onboarding Workflow

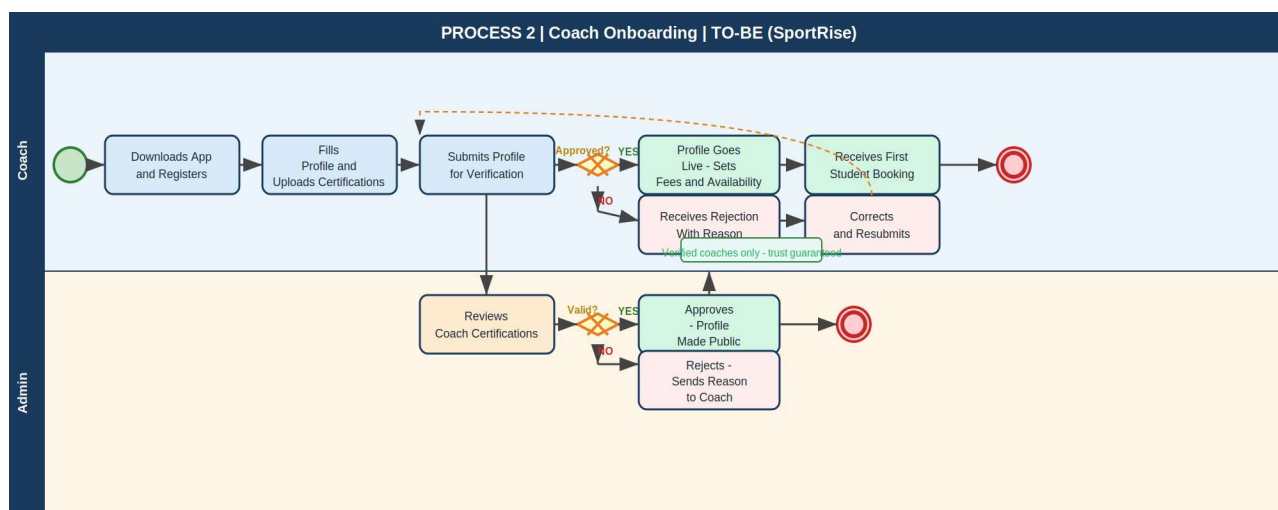


Figure 7.1 — Coach Onboarding Workflow (TO-BE BPMN — Registration through Verification to First Booking)

Tournament Lifecycle Workflow

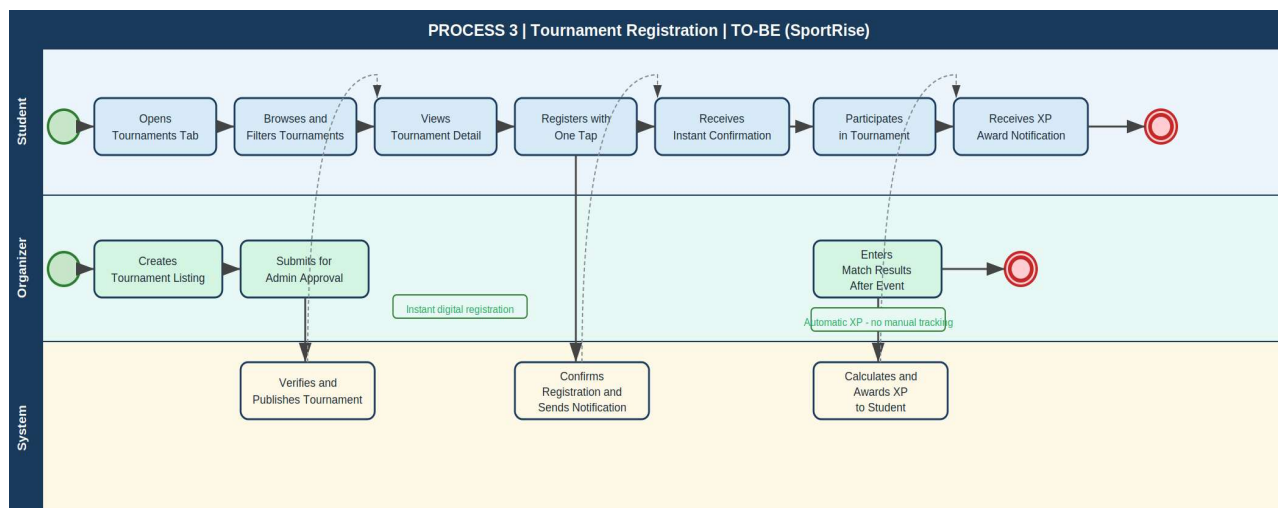


Figure 7.2 — Tournament Lifecycle Workflow (TO-BE BPMN — Listing through Registration to XP Award)

7.4 Approval Processes

Process	Approver	SLA	On Rejection
Coach profile verification	Platform Admin	48 business hours	Coach notified with reason. Can resubmit with corrections.
Tournament listing verification	Platform Admin	24 business hours	Organizer notified with reason. Can resubmit.
Parental consent (under-18)	Parent via OTP	48 hours before expiry	Student account remains inactive. New consent request can be triggered from profile settings.
NP badge award	Automated — no manual approval	Within 60 minutes of trigger	N/A — automated. Admin can review and raise a correction request if false positive suspected.
Scout account activation	Platform Admin	24 business hours	Scout notified. Reapplication with correct credentials required.

8. NON-FUNCTIONAL REQUIREMENTS

8.1 Performance Requirements

NFR ID	Requirement	Target	How Measured
NFR-P01	App screen load time	Under 3 seconds on 4G mid-range Android	Firebase Performance Monitoring
NFR-P02	API response time	Under 2 seconds for all search queries	API response time logs
NFR-P03	XP engine processing	XP updated within 60 seconds of trigger	Timestamp delta: trigger to DB update
NFR-P04	OTP delivery	OTP received within 60 seconds	SMS gateway delivery reports
NFR-P05	Concurrent users	10,000 concurrent users without degradation	Load testing before go-live
NFR-P06	System uptime	99.5% availability — max 3.6 hours downtime per month	AWS CloudWatch monitoring
NFR-P07	Database query time	Under 500ms for all queries with proper indexes	PostgreSQL EXPLAIN ANALYZE

8.2 Security Specifications

NFR ID	Requirement	Implementation
NFR-S01	Data encryption at rest	AES-256 encryption on PostgreSQL database and S3 storage
NFR-S02	Data encryption in transit	TLS 1.3 for all API communications and file transfers
NFR-S03	API authentication	JWT Bearer tokens with 24-hour expiry and refresh token mechanism
NFR-S04	Coach verification gate	API middleware checks is_verified = true before returning coach data in any student-facing endpoint
NFR-S05	Parental consent enforcement	API middleware checks consent_status = confirmed before processing requests from users aged under 18
NFR-S06	Scout anonymity	scout_id is never included in any notification payload or athlete-facing API response
NFR-S07	OTP security	OTPs hashed in database. 10-minute expiry. 3-attempt limit with 30-minute lockout.
NFR-S08	Rate limiting	API rate limiting: 100 requests per minute per IP. OTP endpoint: 3 per hour per mobile number.

8.3 Scalability Needs

- Database: PostgreSQL with read replicas added when concurrent users exceed 5,000
- API: Horizontal scaling via containerization (Docker) — add instances behind load balancer
- File Storage: AWS S3 with CloudFront CDN for profile photo delivery at scale
- Architecture must support adding 10 new sports without structural database changes
- Architecture must support expanding to 5 new cities without code changes — city is a data field

8.4 Compliance Requirements

NFR ID	Regulation	Implementation Requirement
NFR-C01	DPDP Act 2023	Explicit consent required before data collection. Right to deletion must be supported. Privacy policy must be linked during registration.
NFR-C02	Minor Data Protection	Users aged 5-17 require verifiable parental consent via OTP before any platform features are accessible.
NFR-C03	Data Residency	All data stored on servers physically in India (AWS Mumbai ap-south-1 region or equivalent).
NFR-C04	Google Play Policies	App must comply with Play Store developer policies including family policy, data safety form, and content rating.
NFR-C05	Accessibility	WCAG 2.1 AA compliance: minimum 48x48dp touch targets, 4.5:1 text contrast ratio, screen reader support.

9. REPORTING SPECIFICATIONS

9.1 Report Definitions

Report Name	Audience	Description
Platform Health Dashboard	BA, Tech Lead	Real-time: DAU, crash rate, OTP delivery %, session count, error rate
User Growth Report	Product Owner, Sponsor	Weekly new registrations by role, city, sport — trend charts
KPI Progress Report	All Stakeholders	Monthly: all 10 KPIs vs targets with RAG status (Red/Amber/Green)
Coach Performance Report	Admin, BA	Monthly: sessions, avg rating, exceptional count, unique students per coach
XP Distribution Report	BA, Product Owner	Monthly: student count per level, XP histogram, NP badge count
Tournament Success Report	Admin, Organizers	Per tournament: registrations, completion rate, XP awarded, NP triggers
National Prospect Report	Sponsor, Scouts	Monthly: NP badges awarded, scout profile views, recruitment interest sent

Revenue Report	Project Sponsor	Monthly: MRR by stream, cumulative revenue vs projection, break-even tracker
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9.2 Data Sources

Data Source	Data Provided
PostgreSQL — users table	Registration counts, role distribution, city distribution, signup trends
PostgreSQL — student_profiles	XP totals, level distribution, NP badge counts, sport distribution
PostgreSQL — coaching_sessions	Session counts, rating distributions, exceptional rating counts, coach activity
PostgreSQL — xp_transactions	XP event history, XP source breakdown, streak tracking
PostgreSQL — tournaments	Tournament count, sport distribution, venue distribution
PostgreSQL — tournament_registrations	Registration counts per tournament, result distribution, XP from tournaments
Firebase Analytics	DAU, crash rate, screen flow, funnel completion rates, retention
SMS Gateway Logs	OTP delivery rates, failure rates, latency distribution

9.3 Filters and Parameters

Platform Health Dashboard — Key SQL Queries

```
-- Daily Active Users
SELECT DATE(last_login) AS date, COUNT(DISTINCT user_id) AS dau
FROM user_sessions
WHERE last_login >= NOW() - INTERVAL '30 days'
GROUP BY DATE(last_login) ORDER BY date DESC;

-- Level Distribution
SELECT level, COUNT(*) AS students,
       ROUND(100.0*COUNT(*)/SUM(COUNT(*) OVER(), 1) AS pct
FROM student_profiles GROUP BY level ORDER BY level;

-- KPI Progress — NP Badges
SELECT COUNT(*) AS np_badges_awarded
FROM student_profiles WHERE is_national_prospect = true;

-- Revenue MRR
SELECT SUM(fee) AS mrr
FROM subscriptions WHERE status = 'active'
AND period = DATE_TRUNC('month', NOW());
```

9.4 Output Formats

Report Type	Output Format
Platform Health Dashboard	Real-time web dashboard in Admin Panel — auto-refreshes every 5 minutes
All other reports	Power BI interactive dashboards — accessible to authorised stakeholders via link

KPI Progress Report	Power BI + PDF export for stakeholder distribution
Revenue Report	Excel export for finance tracking and investor updates
Tournament Reports	PDF auto-generated and emailed to tournament organizer after event

10. TEST SCENARIOS

10.1 Key Test Cases

TC ID	Test Case	Steps	Expected Result
TC-001	Student registration happy path	Open app → GET STARTED → Student Athlete → Fill form → OTP → Dashboard	Dashboard loads with Level 1 Rookie status and 0 XP in under 5 minutes
TC-002	Age validation - below minimum	Enter age = 4 in registration form → tap Next	Error: 'SportRise is for athletes aged 5 to 30' — cannot proceed
TC-003	Age validation - above maximum	Enter age = 31 in registration form → tap Next	Error: 'SportRise is for athletes aged 5 to 30' — cannot proceed
TC-004	Parental consent trigger	Register with age = 15 → verify OTP	Parental consent screen appears before dashboard — cannot skip
TC-005	Coach discovery - verified only	Open coach discovery → no filters applied	ZERO unverified coaches appear in results under any circumstance
TC-006	Session booking flow	Find coach → view profile → book session → confirm	Booking confirmed, notification sent to both student and coach within 30 seconds
TC-007	XP award - 5/5 exceptional	Coach rates session 5/5 and marks exceptional	Student receives exactly 75 XP within 60 seconds
TC-008	XP award - 4/5 rating	Coach rates session 4/5	Student receives exactly 15 XP within 60 seconds
TC-009	Level up detection	Student at 295 XP receives 10 XP (rating 1-3)	Level updates from 2 to 3, level-up celebration appears, push notification sent
TC-010	NP badge - XP threshold	Student at 1990 XP receives 25 XP (5/5 rating)	NP badge awarded within 60 minutes, scout dashboard updated, student notified
TC-011	NP badge - tournament wins	Student wins 3rd verified tournament	NP badge awarded within 60 minutes regardless of XP level
TC-012	NP badge - distinct coaches	Student receives exceptional from 3 different coaches	NP badge awarded — 3 from same coach does NOT trigger badge
TC-013	Tournament registration - closed	Tap REGISTER on tournament past deadline	Error: 'Registration for this tournament has closed' — cannot register
TC-014	Duplicate tournament registration	Attempt to register for same tournament twice	Error: 'You are already registered for this tournament'
TC-015	Scout search accuracy	Scout filters: Football, Hyderabad, Level 5+	Only football players from Hyderabad at Level 5 or above returned

10.2 Edge Cases

EC ID	Edge Case	Expected Handling
EC-001	Student reaches multiple levels in one XP update	System shows highest new level achieved — celebration for the final level
EC-002	Two NP trigger conditions met simultaneously	Badge awarded once — np_trigger records the first condition met
EC-003	Coach rates session but student app is closed	Push notification sent regardless — celebration shown on next app open
EC-004	OTP requested from Airtel number but goes to BSNL fallback	SMS gateway handles routing — delivery within 60 seconds regardless of operator
EC-005	Student registers for tournament then it gets cancelled	Admin cancels — student receives notification, XP from registration is reversed via compensating transaction
EC-006	Two coaches submit ratings for same student simultaneously	DB transaction isolation ensures correct XP total — both XP amounts added correctly
EC-007	Scout searches with no filters applied	Minimum one filter required — system prompts 'Please apply at least one filter to search'
EC-008	Admin approves coach then coach deletes profile	Coach profile marked inactive — disappears from student search — existing bookings honored
EC-009	Parent gives consent then later requests removal	Parent contacts support — admin can revoke consent, student account suspended pending re-consent
EC-010	NP badge awarded to wrong student due to bug	Admin review dashboard shows NP awards — admin can raise correction request with two-admin approval required

10.3 Integration Test Scenarios

IT ID	Integration Scenario	What is Tested
IT-001	Registration → OTP → Dashboard full flow	End-to-end: form submission, SMS gateway delivery, OTP verification, JWT generation, dashboard load
IT-002	Coach rating → XP update → Level check → NP check → Notification	Full XP Engine pipeline: rating API → XP calculation → DB update → level check → NP evaluation → push notification
IT-003	Tournament registration → result entry → XP award	Full tournament lifecycle: student registers, organizer enters result, system awards XP, updates level
IT-004	Scout search → athlete view → student notification	Cross-module: scout API search → athlete profile fetch → notification service → student push notification
IT-005	Coach verification → coach appears in student search	Admin module → coach verification → search index update → student discovery screen
IT-006	OTP delivery failure handling	SMS gateway returns failure → system retries → fallback mechanism → user-facing error message
IT-007	Concurrent XP updates for same student	Two coaches rate same student simultaneously → both XP amounts correctly added → no race condition

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Functional Design Document | v1.0 | June 2026 | Digital Edify BA & AI Course